

Quadratics in factored form



1 For each of the following equations:

- i Find the y value of the y -intercept.
- ii Find the x values of the x -intercepts.
- iii State the equation of the axis of symmetry.
- iv Find the coordinates of the vertex.
- v Plot the graph of the parabola.

a $y = x(x - 4)$

b $y = x(6 - x)$.

c $y = (x - 5)(x - 3)$

d $y = (x + 4)(x - 2)$

e $y = (1 - x)(x - 5)$

f $y = (5 - x)(3 - x)$.

2 Consider the given equations:

- i Determine the y value of the y -intercept of the graph.
- ii Determine the x values of the x -intercepts.
- iii Complete the table of values for the equation.
- iv Determine the coordinates of the vertex of the parabola.
- v Graph the parabola.

a $y = x(x + 6)$

b $y = (x - 2)(x - 6)$.

x	-5	-4	-3	-2	-1
y					

x	0	2	4	6	8
y					

c $y = (2 - x)(x + 4)$

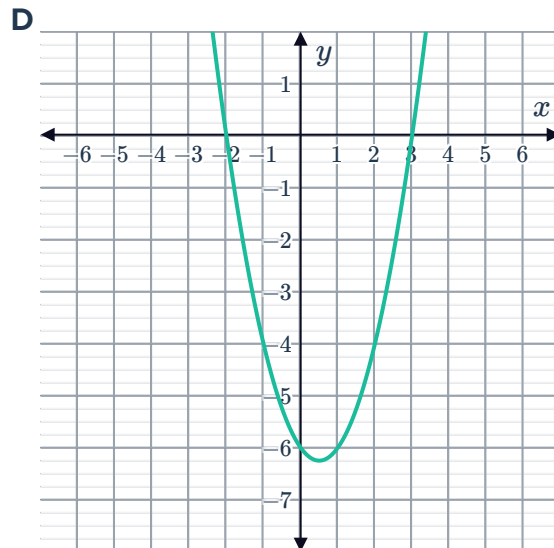
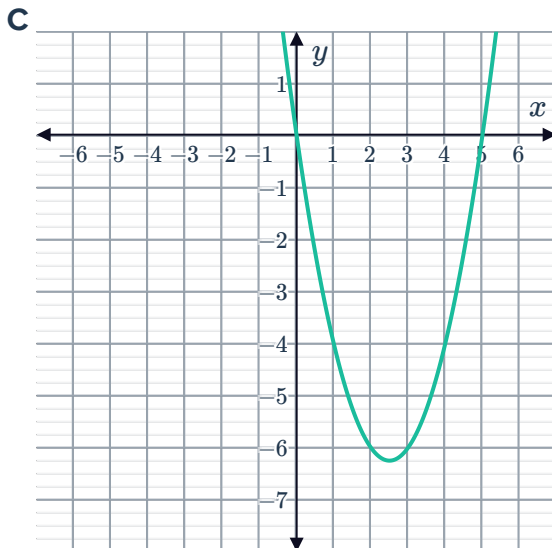
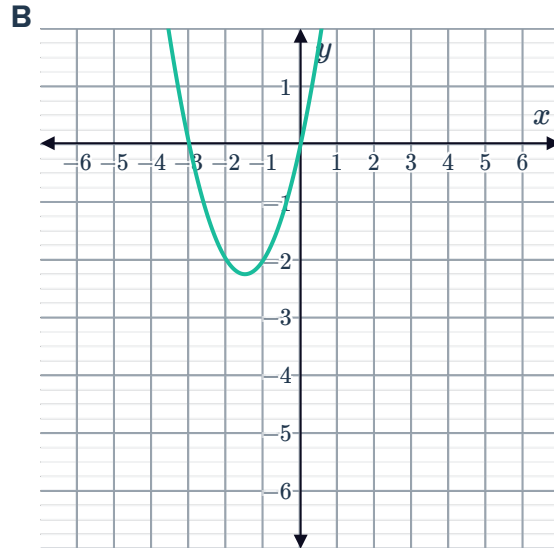
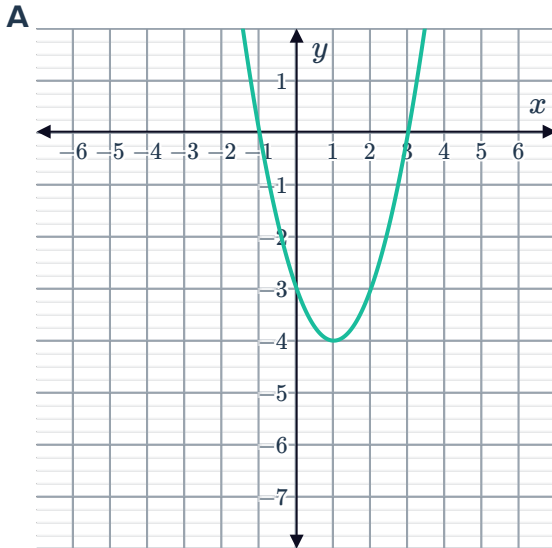
x	-5	-3	-1	1	3
y					

3 Consider the function $y = (x + 5)(x + 1)$.

- a** Plot the x -intercepts and y -intercept of the function, and use these points to plot the graph of $y = (x + 5)(x + 1)$ on the number plane.
- b** Determine the y value of the y -intercept of $y = -(x + 5)(x + 1)$.



4 Consider the following graphs:



Match each of the following equations to the correct graph:

a $y = x(x + 3)$

b $y = x(x - 5)$

c $y = (x + 2)(x - 3)$

d $y = (x - 3)(x + 1)$

5 Consider the equation of a parabola of the form $y = (x - a)(x - b)$.

a The parabola has x -intercepts at $x = -1$ and $x = -5$. Write its equation.

b Determine the y value of the y -intercept of the parabola.

c Use the intercepts to plot the graph of the parabola.



6 Consider a quadratic function of the form $y = a(x - a)(x - b)$.

- a The function has x -intercepts at $x = \sqrt{3}$ and $x = -\sqrt{3}$. Write its equation in standard form.
- b What is the y value of function when $x = 1$?
- c Plot the graph of the function.

7 Consider the following equations:

- i Factor the quadratic.
- ii Find the x values of the x -intercepts.
- iii Find the x value of the vertex.
- iv Find the y value of the vertex.
- v Plot the graph of the function.

a $y = 6x - x^2$ b $y = x^2 + 2x - 8$

8 Consider the equation $y = 4x + x^2$.

- a Factor the expression $4x + x^2$.
- b Find the x values of the x -intercepts.
- c Find the y value of the y -intercept.
- d Complete the table of values for the equation.

x	-4	-3	-2	-1	0
y					

- e Determine the coordinates of the vertex.
- f Graph the function.

9 Consider the equation $y = x^2 + 8x + 12$.

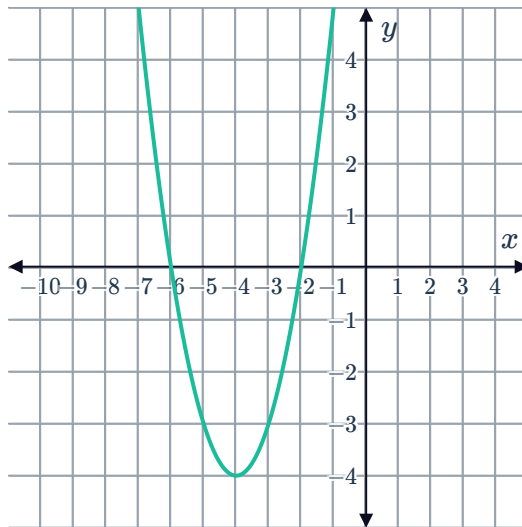
- a Factor the expression $x^2 + 8x + 12$.
- b Solve the x values of the x -intercepts of the quadratic function $y = x^2 + 8x + 12$.

c Complete the table of values for the equation .

x	-8	-6	-4	-2	0
y					

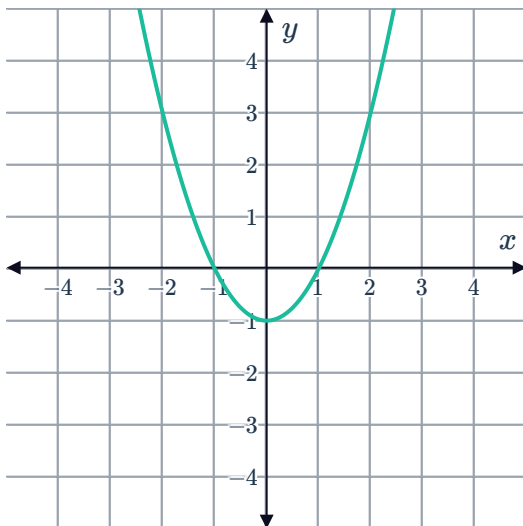
d Determine the coordinates of the vertex.

e Graph the function.

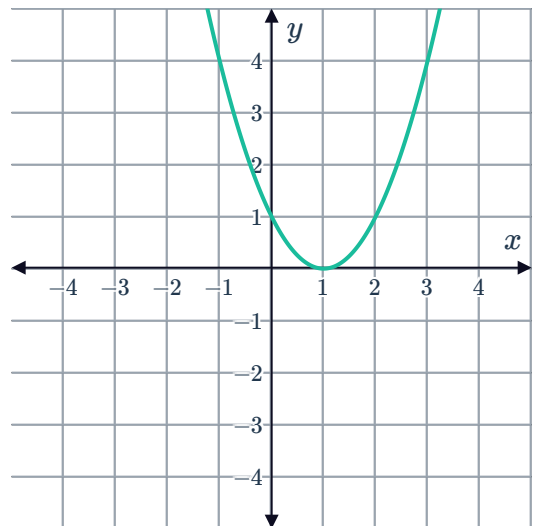


10 State whether the following graphs represent a parabola of the form $y = (x - a)^2$?

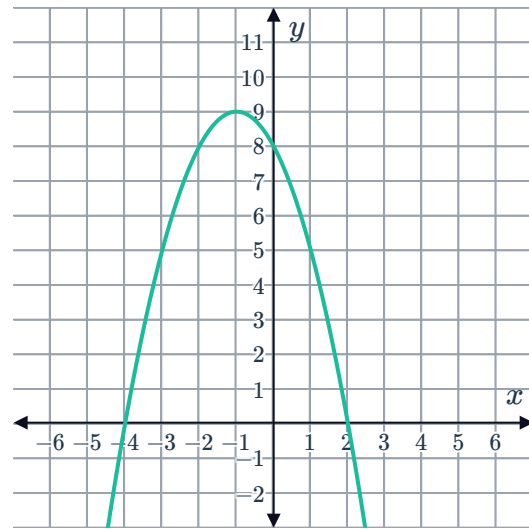
a



b



11 Consider the graph.



- a State the y value of the y -intercept of the graph.
- b Write the equation of this graph in factored form.

12 State the axis of symmetry of the parabola $y = k(x - 7)(x + 7)$ for any value of k .

13 A parabola has its vertex at $x = -5$ and one of the x -intercepts is $x = 1$.

- a Find the x value of the other x -intercept?
- b If it has a y -intercept at -11 , determine the equation of the parabola.
- c Find the coordinates of the vertex.

14 Consider the equation $y = (x - b)(x + b)$.

- a Write the equation in standard form.
- b Find the value of a such that the parabola $y = x^2 + ax - 9$ will have x -intercepts that are equidistant from the origin.

15 A football is kicked into the air and its height h meters above the ground at time t seconds after being kicked is given by $h = -t^2 + 20t$.

- a Assuming the ball starts at height 0, determine the time t the ball will hit the ground.
- b Find the greatest height the ball reaches above the ground.

Additional questions

16 State the factored form for a quadratic equation and describe its features.

17 Determine the equation of the parabola that has its vertex at $(-2, 5)$ and passes through the point $(0, 1)$.



18 For each of the following quadratic equations, state the x -values of the x -intercepts:

a $y = (x - 1)(x - 4)$

b $y = (x - 3)(x + 7)$

c $y = x(x - 4)$

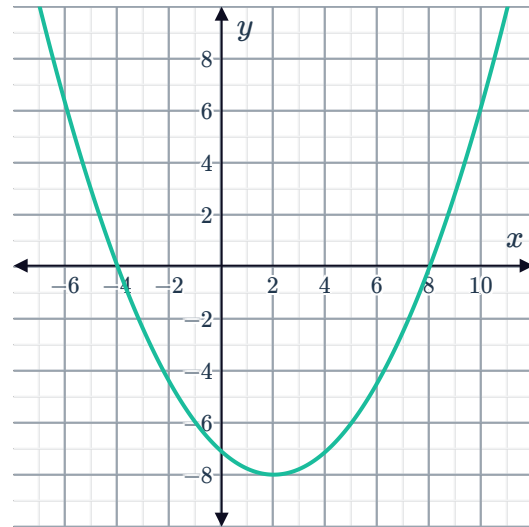
d $y = 2(x - 1)(x + 5)$

19 For each of the following quadratic functions, calculate the y -value of the y -intercept:

a $y = 3(x - 1)(x - 8)$

b $y = x(x + 9)$

20 Identify the axis of symmetry of this quadratic function.



21 For each quadratic function:

i State the coordinates of the x -intercepts.

ii Determine the axis of symmetry.

iii Determine the coordinates of the vertex.

iv Draw the graph of the function.

a $y = (x - 2)(x + 4)$

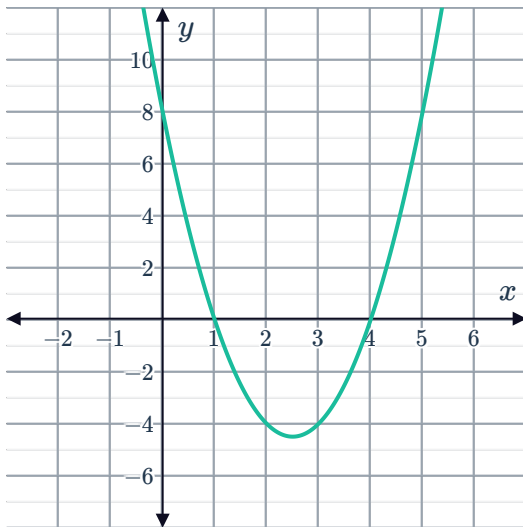
b $y = -2(x - 1)(x - 5)$

22 For each graph:

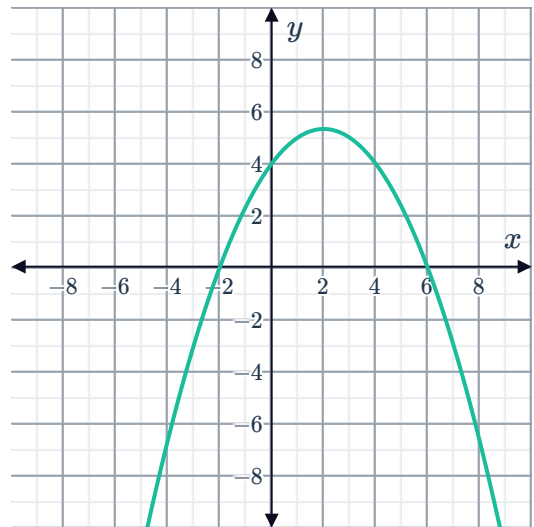


- i Identify the x -values of the x -intercepts.
- ii Identify the y -value of the y -intercept.
- iii Write the equation of the quadratic function in factored form.

a



b

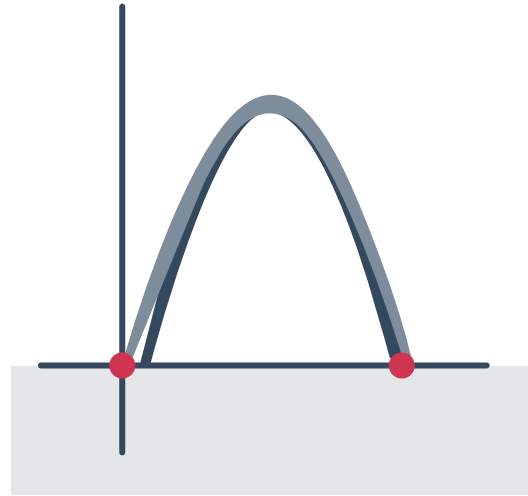


23 A quadratic function has x -intercepts of $(5, 0)$ and $(-3, 0)$, and a y -intercept of $(0, 60)$.
Write the equation of the function in factored form.

24 A quadratic function passes through the points $(-2, 0)$, $(9, 0)$ and $(0, -6)$.
Write the equation of the function in factored form.

25 The x -values of the x -intercepts of a quadratic function are $x = -4$ and $x = -7$. The graph of the function passes through the point $(-3, 12)$.
Write the equation of the function in factored form.

- 26** Xia notices that the Sunshine State Arch is in the shape of a quadratic function. They know that the arch has a height of 110 ft and the feet of the arch are 100 ft apart. Xia chooses to let the x -intercepts of the arch be the origin and $(100, 0)$.



- a** Determine the coordinates of the vertex of the arch.
 - b** Write the equation of the quadratic function for Xia's representation of the Sunshine State Arch in factored form.
 - c** Find the domain and range of the Sunshine State Arch.
- 27** Ami throws a javelin forwards in a parabolic arc from the ground. Using photos that her friend Maryellen is taking from the stands, she determines that 10 yards horizontally from where she threw the javelin, it reaches a maximum height 5 yards above the ground. Ami models her throw with the origin of a cartesian plane being the point 20 yards behind her.
- a** For Ami's model, state the coordinates of the x -intercepts.
 - b** Write the equation for the quadratic function modelling the path of Ami's throw in factored form.
- 28** Wilson tosses an eraser into the air and counts how long it takes for the eraser to return to his hand. He estimates that it takes 2 seconds and that he is tossing the eraser 6 ft into the air. Wilson models the height of the eraser above his hand in feet as a function h of time t in seconds, letting $t = 0$ be when he tosses the eraser and $h = 0$ be the height of his hand.
- a** Explain how to find the intercepts of the function and state them.
 - b** Assuming that the path of the eraser is symmetric going up and coming down, state the coordinates of the vertex of the function.
 - c** Draw a graph of the function.
 - d** Write the equation of the function in factored form.



- 29 A quadratic function has the factored form equation $y = k(x - x_0)(x - 12)$. If the function has a vertex at the point $(8, 15)$, determine the values of x_0 and k . Justify your answers.